Student Name: Shih-Yu Yang **UIN:** 439000331

1. **Setup :** NVIDIA A100-PCIE-40GB GPU, execution within a Conda environment on TAMU Grace HPC. Total 3DGS optimization time: ~6 minutes 28 seconds (30,000 iterations).

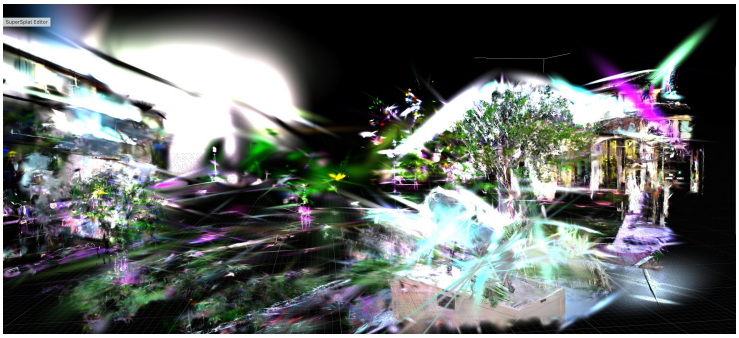
2. **Data Split (1-in-8 Protocol):** Total dataset contains 20 multi-view images. Following the 1-in-8 evaluation rule, 3 images (0,8,16) are held out for testing, while the remaining 17 images are used for training.

3. **Quantitative Results**

Metric	My Result	Interpretation
PSNR	21.10 dB	Higher is better
SSIM	0.799	Higher is better
LPIPS	0.207	Lower is better

4. **Global Reconstruction View**

Figure 1: Global view of the reconstructed 3D Gaussian scene inspected in the 3DGS viewer, showing overall geometry and spatial distribution of Gaussian ellipsoids.



5. **Novel-View Comparison (Held-out Test Camera)**

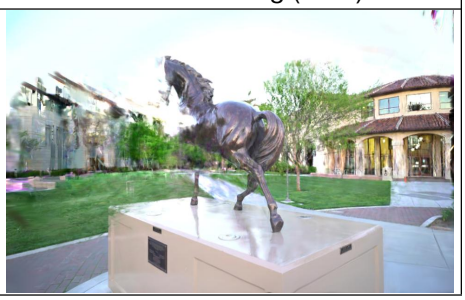
Ground Truth (Test View)	3DGS Rendering (Ours)
	

Figure 2: Side-by-side comparison between the held-out ground-truth viewpoint and the novel-view renderinggenerated by 3D Gaussian Splatting at 30,000 iterations.

6. **Brief Analysis**

The 3D Gaussian Splatting pipeline successfully reconstructs the primary global structure of the statue, reaching a test PSNR of 21.10 dB and SSIM of 0.799. However, noticeable artifacts appear in the novel-view rendering, including structural loss where part of the horse head disappears and fine details blur. Additionally, prominent needle-like Gaussian distortions are visible in the global reconstruction view. These failures occur because the model is trained on only 17 sparse images, leaving camera rays under-constrained and causing the adaptive density control to stretch Gaussians into needle-like shapes along unverified depths.